Identification of the Worst-case Static Voltage Stability Margin Interval of AC/DC Power System Considering Uncertainty of Renewables

Wanbin Liu, Shunjiang Lin, *Senior Member*, *CSEE*, *Senior Member*, *IEEE*, Yuerong Yang, Mingbo Liu, *Senior Member*, *CSEE*, and Qifeng Li, *Senior Member*, *IEEE*

***Abstract*—Calculation of static voltage stability margin (SVSM) of AC/DC power systems with lots of renewable energy sources (RESs) integration requires consideration of uncertain load growth and renewable energy generation output. This paper presents a bi-level optimal power flow (BLOPF) model to identify the worst-case SVSM of an AC/DC power system with line commutation converter-based HVDC and multi-terminal voltage sourced converter-based HVDC transmission lines. Constraints of uncertain load growth's hypercone model and control mode switching of DC converter stations are considered in the BLOPF model. Moreover, uncertain RES output fluctuations are described as intervals, and two three-level optimal power flow (TLOPF) models are established to identify interval bounds of the system worst-case SVSM. The two TLOPF models are both transformed into max–min bi-level optimization models according to independent characteristics of different uncertain variables. Then, transforming the inner level model into its dual form, max–min BLOPF models are simplified to single-level optimization models for direct solution. Calculation results on the modified IEEE-39 bus AC/DC case and an actual large-scale AC/DC case in China indicate correctness and efficiency of the proposed identification method.**

***Index Terms*—AC/DC power system, control mode switching, convex relaxation, dual programming, multi-level optimization, optimal power flow, static voltage stability margin.**

NOMENCLATURE

A. Sets and Indices

N_{G}	Number of generators.
N_{RE}	Number of renewable energy stations (RESs).
N_{LCC}	Number of LCC-HVDC stations.
N_{VSC}	Number of VSC-HVDC stations.
n	Number of buses.

B. Parameters

G_{ij}, B_{ij}	Mutual conductance and susceptance between buses i and j.
$P_{\mathrm{L}i0}, Q_{\mathrm{L}i0}$	Initial active and reactive load power.
$V_{i\max}, V_{i\min}$	Upper and lower bounds of V_i.
$P_{\mathrm{G}i\max}, P_{\mathrm{G}i\min}$	Upper and lower bounds of $P_{\mathrm{G}i}$.
$Q_{\mathrm{G}i\max}, Q_{\mathrm{G}i\min}$	Upper and lower bounds of $Q_{\mathrm{G}i}$.
$P_{\mathrm{G}i0}$	Initial active generator output.
$V_{\mathrm{G}i\mathrm{ref}}$	Reference value of the ith generator's terminal voltage.
R_i	Ramp rate of the ith normal generator.
$K_{\mathrm{p}i}$	Number of polar pair of the DC line.
$K_{\mathrm{d}i}, X_{ci}$	Transformer ratio and commutation reactance of the LCC-HVDC station of bus i.
R_{dc}	DC line resistance.
μ_i	DC voltage utilization.
$g_{il,\mathrm{d}}$	Conductance between DC circuit bus i and bus j.
Y_i, α_i	Equivalent admittance and impedance angle of VSC-HVDC converter station of bus i.
$U_{\mathrm{a}i\max}, U_{\mathrm{a}i\min}$	Upper and lower bounds of auxiliary variable $U_{\mathrm{a}i}$.
$U_{\mathrm{b}i\max}, U_{\mathrm{b}i\min}$	Upper and lower bounds of auxiliary variable $U_{\mathrm{b}i}$.

C. Variables

λ	Load growth parameter.
$b_{\mathrm{PL}i}$, $b_{\mathrm{QL}i}$	Active and reactive load growth direction.
$P_{\mathrm{L}i}$, $Q_{\mathrm{L}i}$	Active and reactive load power.
$P_{\mathrm{G}i}$, $Q_{\mathrm{G}i}$	Active and reactive generator output.
ϕ_i	Multiple of the current load growth direction deviating from $P_{\mathrm{L}i0}$.
β_i	Given uncertain degree of load growth direction.
$P_{\mathrm{RE}i}$, $Q_{\mathrm{RE}i}$	Active and reactive RES output.
$P_{\mathrm{RE}i0}$	Central value of the active RES output interval.
V_i/V_j	Bus voltage.

Manuscript received September 19, 2022; revised March 9, 2023; accepted March 29, 2023. Date of online publication September 8, 2023; date of current version October 18, 2023. This work was supported by the National Natural Science Foundation of China (Grant No. 51977080), and the Natural Science Foundation of Guangdong Province (Grant No. 2022A1515010332). Q. F. Li's work is supported by the U.S. National Science Foundation (Grant #2124849).

W. B. Liu, S. J. Lin (corresponding author, email: linshj@scut.edu.cn; ORCID: https://orcid.org/0000-0001-8201-2045), Y. R. Yang and M. B. Liu are with the School of Electric Power Engineering, South China University of Technology, Guangzhou 510640, China, and the Guangdong Key Laboratory of Clean Energy Technology, South China University of Technology, Guangzhou 511458, China.

Q. F. Li is with the Department of Electrical and Computer Engineering, University of Central Florida, Orlando, FL 32816, USA.

DOI: 10.17775/CSEEJPES.2022.06390

2096-0042 © 2022 CSEE. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

θ_{ij}	Voltage phase difference between buses i and j.
$U_{\mathrm{d}i}/I_{\mathrm{d}i}$	DC voltage and current of converter station of bus i.
$U_{\mathrm{dR}i}/U_{\mathrm{dI}i}$	DC voltage of rectifier/inverter side.
$\phi_{\mathrm{d}i}$	Power factor angle of LCC-HVDC station of bus i.
ξ_i	Extinction angle of LCC-HVDC station of bus i.
$P_{\mathrm{vd}i}/Q_{\mathrm{vd}i}$	Active and reactive power absorbed by VSC-HVDC station from AC system.
M_i	Modulation ratio of VSC-HVDC station.
δ_i	Phase angle difference between V_i and $U_{\mathrm{d}i}$.
$b_{\mathrm{PG}i}$	Growth direction of the ith generator output.
$V_{\mathrm{G}i}$	Voltage of the ith generator's terminal.
$V_{\mathrm{a}i}$, $V_{\mathrm{b}i}$	Auxiliary variables used to correct the terminal voltage. of the ith generator when the reactive output reaches its lower/upper bound.
$\boldsymbol{x}$	Vector of the variables of inner level optimization model.

I. Introduction

NOWADAYS, LCC-HVDC and multi-terminal voltage sourced converter-based HVDC (MTVSC-HVDC) transmission technologies are widely used in actual power grids [1]–[3]. For example, in China Southern Power Grid, an ±800 kV/8000 MW 'Kun-Liu-Long' ultra-high-voltage MTVSC-HVDC transmission system was put into operation, and AC/DC power grid now includes eight AC and eleven DC transmission channels [4]. Different operation characteristics and control modes of the HVDC transmission system will have significant impact on the operation state of the entire AC/DC power system and will also affect static voltage stability degree of the system. In addition, with many renewable energy generation stations connected to the power grid, uncertain fluctuation of their power outputs will result in uncertain fluctuation of the system operation state, which brings great challenges to maintain voltage stability of the AC/DC power system [5], [6]. Static voltage stability margin (SVSM) is a commonly used index for evaluating static voltage stability degree of a power system, which represents maximum load growth the system can bear in the current operation state [5]–[7]. Therefore, it is urgent to propose an SVSM calculation method for AC/DC power systems with LCC-HVDC and MTVSC-HVDC considering uncertain renewable energy output.

Recently, several publications have focused on SVS analysis of AC/DC power systems. Reference [8] analyzed the influence of VSC-HVDC on LCC-HVDC at the same bus, and it also analyzed the SVS of the system, including LCC-HVDC or VSC-HVDC, with maximum available power as the index. In [9], a single-infeed model is derived to represent a multi-infeed DC transmission system, and the effective short-circuit ratio is applied to voltage stability analysis of a single-infeed HVDC system was extended to analyze stability of a multi-infeed HVDC system. Reference [10] proposed a general SVS analysis method, deduced a general voltage stability index called the hybrid generalized short circuit ratio, and solved voltage stability critical point of the hybrid multi-infeed DC system. Reference [11] proposed a control algorithm based on the VSC-HVDC system to improve SVS, which showed VSC control input will affect the local and global SVSM. Based on the aforementioned studies, it can be concluded establishing an accurate steady-state model of LCC-HVDC and VSC-HVDC converter stations is helpful to reflect power flow distribution of the system more accurately. However, in SVS analysis of the aforementioned studies, the control mode of HVDC was fixed and did not take into account the control mode of HVDC will change with load growth in the system [12], [13]. Switching of the HVDC control mode will affect calculation results of the SVSM of an AC/DC power system, so it is necessary to model switching of control modes in SVSM calculation.

For a power system integrating numerous RESs, if the uncertainty of RES output is considered, SVSM of the system will be an uncertain quantity instead of a definite value [14]. Interval method is widely used for uncertain analysis because it only needs to give RES output range in advance without excessive sampling, and RES output range is easy to obtain according to historical operation data. In [15], [16], interval number was used to describe uncertainty of RES output, and corresponding SVSM interval was calculated by the affine interval algorithm and continuous power flow method. Reference [17] proposed the interval optimization algorithm and the OPF method to calculate the SVSM interval considering uncertain RES output. However, aforementioned studies only investigated calculation of SVSM interval under a fixed load growth direction (LGD), i.e., initial LGD, and did not consider uncertainty of LGD in actual operation of the power grid. There are two main methods for considering uncertainty of LGD. One is to calculate SVSM corresponding to various possible LGDs to form a boundary of the SVS region, as shown in [18]. Another is to analyze the worst case of the system considering uncertain LGD to assess SVS, which imposes less computation burden. References [19], [20] proposed a calculation method for the local closest bifurcation point to the current operation point. Minimum SVSM from the load power parameter space to the local closest bifurcation point was used as the voltage collapse index. Considering uncertain LGD, [21] introduced the calculation method for the worst-case SVSM. However, the aforementioned studies only applied the proposed methods to pure AC power systems and did not consider uncertain RES output. Therefore, how to calculate the worst-case SVSM interval of an AC/DC power system considering uncertain RES output and the uncertain LGD still requires further research.

The contributions of this paper are as follows: 1) A method to identify the worst-case SVSM interval of an AC/DC power system with LCC-HVDC and MTVSC-HVDC transmission lines considering control mode switching of DC converter stations is proposed, along with the hypercone model to describe uncertain LGD. 2) Two TLOPF models are constructed to solve two bounds of the worst-case SVSM interval of an AC/DC power system, and a method to transform TLOPF models into single-level convex optimization models for solution is proposed. The proposed method has much higher computational efficiency than the Monte Carlo or Latin

hypercube sampling method, which can be applied to the worst-case SVSM identification in actual large-scale AC/DC systems accurately and efficiently.

The rest of this paper is organized as follows. Section II presents worst-case SVSM identification model for AC/DC power systems based on the OPF method and two TLOPF models for worst-case SVSM interval identification. Section III proposes the solution method for three-level optimization models. Section IV details case studies on a modified IEEE-39 bus system and an actual 8733 bus AC/DC power grid in China. Section V presents the conclusions.

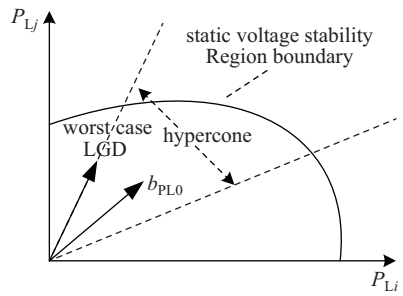


Fig. 1. Load growth hypercone model and worst-case LGD.

II. The Worst-case SVSM Interval Identification Model for AC/DC Power Systems

A. Identification Model for the Worst-case SVSM Based on OPF Method

In traditional SVSM calculation, the LGD can be expressed as (1), which assumes the load of each bus increases proportionally according to its initial load, that is, $b_{\mathrm{PL}i} = P_{\mathrm{L}i0}$ and $b_{\mathrm{QL}i} = Q_{\mathrm{L}i0}$ [14]–[17]. Obtained SVSM corresponding to the SVS limit point is generally called SVSM under the initial LGD.

$$\begin{cases} P_{\mathrm{L}i} = P_{\mathrm{L}i0} + \lambda b_{\mathrm{PL}i} \\ Q_{\mathrm{L}i} = Q_{\mathrm{L}i0} + \lambda b_{\mathrm{QL}i} \end{cases} \tag{1}$$

However, in actual power system operation, the LGD has uncertainty and may deviate from initial LGD. Only calculating SVSM under the initial LGD will be too optimistic and bring unsecure operation risks to the power system. Thus, it is necessary to calculate SVSM under worst-case LGD [19]–[21]. According to [21], uncertain LGD from the current operation point can be described as a hypercone that includes all future operation point trajectories. The hypercone starts with the current operating point, and uncertainty of future LGD is taken as the "thickness" of the hypercone. This model assumes the future operating point is in the hypercone, that is, future LGD changes will be within the hypercone range. In this paper, ϕ is introduced to represent deviation degree of actual LGD from initial LGD, and the hypercone model of the uncertainty of LGD is described as (2a)–(2c). To simplify discussion, it is assumed the uncertain deviation degrees of LGD of all the load buses are the same, i.e., their β_i values are equal. In addition, (2c) indicates each load bus maintains a constant power factor during the load growth process.

$$b_{\mathrm{PL}i} = P_{\mathrm{L}i0}\phi_i \tag{2a}$$

$$1 - \beta_i \le \phi_i \le 1 + \beta_i \tag{2b}$$

$$b_{\mathrm{QL}i} = b_{\mathrm{PL}i} Q_{\mathrm{L}i0} / P_{\mathrm{L}i0} \tag{2c}$$

In the hypercone range, there is an LGD that will result in minimum total load growth when power system reaches SVS limit, i.e., corresponding SVSM is smallest. Since total load growth the power system can withstand under this LGD is smallest, system will most easily reach the SVS limit from current operating point along this LGD. Thus, this LGD is called worst-case LGD, and SVSM under worst-case LGD is called worst-case SVSM, as shown in Fig. 1.

Considering variation of LGD, $b_{\mathrm{PL}i}$ is not equal to initial load $P_{\mathrm{L}i0}$. Thus, λ cannot represent total load growth of SVS limit point relative to initial load $P_{\mathrm{L}i0}$, so it cannot be directly used as SVSM of the system. When LGD changes, actual load growth of each bus at SVS limit point can be obtained. Hence, the SVSM can be represented by total load growth from current operating point to the SVS limit point.

Thus, based on the OPF method and hypercone model of load growth, the objective function of established bi-level OPF model is equation (3).

$$\max_{\phi} \min_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \tag{3}$$

Equation (4) is pure AC buses' power balance equation. For buses without RES injection, $P_{\mathrm{RE}i} = Q_{\mathrm{RE}i} = 0$; for buses not connected to normal generators, $P_{\mathrm{G}i} = Q_{\mathrm{G}i} = 0$. Assuming RESs operate with a constant power factor, then $Q_{\mathrm{RE}i} = P_{\mathrm{RE}i} * tan\phi_{\mathrm{RE}i}$. Equations (5) and (6) are power balance equations of AC buses with LCC/VSC-HVDC stations' connection, respectively. The $\pm$ sign in (5a) is negative for rectifier station and positive for inverter station.

$$\begin{aligned} &P_{\mathrm{G}i} + P_{\mathrm{RE}i} - (P_{\mathrm{L}i0} + \lambda P_{\mathrm{L}i0}\phi_i) \\ &\quad - V_i \sum_{j=1}^{n} V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) = 0 \end{aligned} \tag{4a}$$

$$\begin{aligned} &Q_{\mathrm{G}i} + Q_{\mathrm{RE}i} - (Q_{\mathrm{L}i0} + \lambda Q_{\mathrm{L}i0}\phi_i) \\ &\quad - V_i \sum_{j=1}^{n} V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) = 0 \end{aligned} \tag{4b}$$

$$- V_i \sum_{j=1}^{n} V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \pm K_{\mathrm{p}i} U_{\mathrm{d}i} I_{\mathrm{d}i} = 0 \tag{5a}$$

$$\begin{aligned} &- V_i \sum_{j=1}^{n} V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) \\ &\quad - K_{\mathrm{p}i} U_{\mathrm{d}i} I_{\mathrm{d}i} \tan\phi_{\mathrm{d}i} = 0 \end{aligned} \tag{5b}$$

$$- V_i \sum_{j=1}^{n} V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) - P_{\mathrm{vd}i} = 0 \tag{6a}$$

$$- V_i \sum_{j=1}^{n} V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) - Q_{\mathrm{vd}i} = 0 \tag{6b}$$

Equations (7) and (8) are operation characteristics con-

straints of two types of HVDC transmission lines, respectively.

$$U_{\mathrm{d}i} = 3\sqrt{2}/\pi \cdot V_i K_{\mathrm{d}i} \cos\xi_i - 3/\pi \cdot X_{\mathrm{c}i} I_{\mathrm{d}i} \tag{7a}$$

$$\tan^2\phi_{\mathrm{d}i} = (3\sqrt{2}/\pi \cdot V_i K_{\mathrm{d}i}/U_{\mathrm{d}i})^2 - 1 \tag{7b}$$

$$U_{\mathrm{dR}i} = U_{\mathrm{dI}i} + I_{\mathrm{d}i} R_{\mathrm{dc}} \tag{7c}$$

$$P_{\mathrm{vd}i} + \mu_i M_i/\sqrt{2} \cdot V_i U_{\mathrm{d}i} Y_i \cos(\delta_i + \alpha_i) - V_i^2 Y_i \cos\alpha_i = 0 \tag{8a}$$

$$Q_{\mathrm{vd}i} + \mu_i M_i/\sqrt{2} \cdot V_i U_{\mathrm{d}i} Y_i \sin(\delta_i + \alpha_i) - V_i^2 Y_i \sin\alpha_i = 0 \tag{8b}$$

$$U_{\mathrm{d}i} I_{\mathrm{d}i} - \mu_i M_i/\sqrt{2} \cdot V_i U_{\mathrm{d}i} Y_i \cos(\delta_i - \alpha_i) + (\mu_i M_i/\sqrt{2} \cdot U_{\mathrm{d}i})^2 Y_i \cos\alpha_i = 0 \tag{8c}$$

$$I_{\mathrm{d}i} - \sum_{l=1}^{m} g_{il,\mathrm{d}} U_{\mathrm{d}l} = 0 \tag{8d}$$

Equation (9) describes growth mode of active power output of generators. Except for swing bus generator, normal generators assign total active load power growth according to ratio of their active power margin to total active power margin; hence, $b_{\mathrm{PG}i}$ is described as (10). Equations (11a)-(11e) are complementary constraints of $Q_{\mathrm{G}i}$ and $V_{\mathrm{G}i}$.

$$P_{\mathrm{G}i} = P_{\mathrm{G}i0} + \lambda b_{\mathrm{PG}i} \tag{9}$$

$$b_{\mathrm{PG}i} = \frac{P_{\mathrm{G}i\max} - P_{\mathrm{G}i0}}{\sum_{j=1}^{N_{\mathrm{G}}-1}(P_{\mathrm{G}j\max} - P_{\mathrm{G}j0})} \sum_{k=1}^{n} b_{\mathrm{PL}k} \tag{10}$$

$$Q_{\mathrm{G}i\min} \le Q_{\mathrm{G}i} \le Q_{\mathrm{G}i\max} \tag{11a}$$

$$(Q_{\mathrm{G}i} - Q_{\mathrm{G}i\min}) V_{\mathrm{a}i} = 0 \tag{11b}$$

$$(Q_{\mathrm{G}i} - Q_{\mathrm{G}i\max}) V_{\mathrm{b}i} = 0 \tag{11c}$$

$$V_{\mathrm{G}i} = V_{\mathrm{G}i\mathrm{ref}} + V_{\mathrm{a}i} - V_{\mathrm{b}i} = 0 \tag{11d}$$

$$V_{\mathrm{a}i} \ge 0, V_{\mathrm{b}i} \ge 0 \tag{11e}$$

Based on OPF method, bi-level worst-case SVSM identification model of an AC/DC power system with LCC/VSC-HVDC stations is established as follows:

$$\max_{\phi} \min_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0} \phi_i$$
$$\text{s.t.} \quad \text{(2b), (4)–(11)} \tag{12}$$

In actual operation, control modes of the two types of HVDC transmission lines need to be given. For the LCC-HVDC transmission line, control mode of constant $I_{\mathrm{d}i}$ and $U_{\mathrm{dR}i}$ is adopted. Thus, inverter side DC voltage $U_{\mathrm{dI}i}$ can be obtained by (7c). That is, once control mode is given, $U_{\mathrm{d}i}$ and $I_{\mathrm{d}i}$ of the LCC-HVDC station are known, and operation characteristic constraints of LCC-HVDC transmission line in the OPF model is simplified to (7a)–(7b). For the multi-terminal VSC-HVDC transmission line, control mode of constant $U_{\mathrm{d}i}$ and $Q_{\mathrm{vd}i}$ for rectifier station and constant $I_{\mathrm{d}i}$ and $Q_{\mathrm{vd}i}$ for inverter station is adopted [22]. Thus, other DC variables can be obtained by (8d). Hence, operation characteristic constraints of multi-terminal VSC-HVDC transmission line in the OPF model is simplified to (8a)–(8c).

As system loads increase, AC bus voltage of inverter side of the LCC-HVDC system will decrease. If control mode is constant DC voltage, to keep $U_{\mathrm{dI}i}$ unchanged, inverter control angle $\xi_{\mathrm{I}i}$ is required to decrease. To prevent commutation failure of converters, $\xi_{\mathrm{I}i}$ has a minimum limit value $\xi_{\mathrm{I}i\min}$ [23]. Therefore, when $\xi_{\mathrm{I}i}$ is decreased to minimum limit value $\xi_{\mathrm{I}i\min}$, to ensure normal commutation of converters, it is necessary to change control mode to constant minimum control angle of the inverter. Hence, $U_{\mathrm{dI}i}$ will be reduced. Equation (7c) indicates under constant DC current control mode, variations of $U_{\mathrm{dR}i}$ and $U_{\mathrm{dI}i}$ will be the same. Added constraints representing control mode switching of the LCC-HVDC converter station with load growth in the OPF model are expressed in (13a)–(13d).

$$(\cos\xi_{\mathrm{I}i\,\min} - \cos\xi_{\mathrm{I}i}) V_{\mathrm{d}i} = 0 \tag{13a}$$

$$U_{\mathrm{dR}i} = U_{\mathrm{dR}i\mathrm{ref}} - V_{\mathrm{d}i} \tag{13b}$$

$$U_{\mathrm{dI}i} = U_{\mathrm{dI}i\mathrm{ref}} - V_{\mathrm{d}i} \tag{13c}$$

$$V_{\mathrm{d}i} \ge 0 \tag{13d}$$

Moreover, as system loads increase, AC bus voltage of the VSC-HVDC system will decrease, especially for inverter side bus under a heavy load. To prevent AC bus voltage of inverter side of VSC-HVDC system from exceeding secure operation limit, it is necessary to change control mode from constant AC reactive power $Q_{\mathrm{vd}i}$ to constant AC bus voltage of inverter side. Hence, $Q_{\mathrm{vd}i}$ will be reduced. Added constraints representing control mode switching of VSC-HVDC converter station with load growth in OPF model are expressed in (14a)–(14c).

$$(V_i - V_{i\min}) Q_{ei} = 0 \tag{14a}$$

$$Q_{\mathrm{vd}i} = Q_{\mathrm{vd}i\mathrm{ref}} - Q_{ei} \tag{14b}$$

$$Q_{ei} \ge 0 \tag{14c}$$

Therefore, the proposed bi-level OPF model for calculating worst-case SVSM of an AC/DC power system considering control mode switching includes (2b), (3)–(6), (7a), (7b), (8a), (8b), (9)–(11) and (13)–(14).

B. The Worst-case SVSM Interval Identification Model Considering Uncertain RES Output

With high penetration of renewable energy into power systems, uncertain fluctuation of RES output will be large. Swing bus generator will not be able to bear such a large RES power fluctuation, so other generators will be required to bear uncertain RES power fluctuation. In order to adjust generator output to eliminate frequency deviation caused by uncertain fluctuation of RES output, the strategy of allocating required total power output fluctuation to units based on their ramp rates is used in most automatic generation control (AGC) systems [24]. Therefore, an additional term $\Delta P_{\mathrm{G}i}$ should be added to (9) to describe power output increment of ith generator caused by uncertain fluctuation of RES output, as (15)–(16).

$$P_{\mathrm{G}i} = P_{\mathrm{G}i0} + \lambda b_{\mathrm{PG}i} + \Delta P_{\mathrm{G}i} \tag{15}$$

$$\Delta P_{\mathrm{G}i} = -\frac{R_i}{\sum_{j=1}^{N_{\mathrm{G}}-1} R_j} \sum_{i=1}^{N_{\mathrm{RE}}} (P_{\mathrm{RE}i} - P_{\mathrm{RE}i0}) \tag{16}$$

Since uncertain RES output is described as an interval, corresponding worst-case SVSM of system is also an interval

instead of a definite value. Thus, calculating worst-case SVSM interval means calculating its lower and upper bounds. The two bounds of the worst-case SVSM interval of an AC/DC power system considering uncertain RES output can be determined using two TLOPF models, expressed in (17) for lower bound, and (18) for upper bound.

$$\begin{cases} \max\limits_{\boldsymbol{P}_{\mathrm{RE}}} \max\limits_{\phi} \min\limits_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \\ \text{s.t.} \begin{cases} \text{(2b), (4)–(6), (7a), (7b), (8a), (8b),} \\ \quad \text{(10)–(11), (13)–(16)} \\ P_{\mathrm{RE}i\min} \le P_{\mathrm{RE}i} \le P_{\mathrm{RE}i\max} \end{cases} \end{cases} \tag{17}$$

$$\begin{cases} \min\limits_{\boldsymbol{P}_{\mathrm{RE}}} \max\limits_{\phi} \min\limits_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \\ \text{s.t.} \begin{cases} \text{(2b), (4)–(6), (7a), (7b), (8a), (8b),} \\ \quad \text{(10)–(11), (13)–(16)} \\ P_{\mathrm{RE}i\min} \le P_{\mathrm{RE}i} \le P_{\mathrm{RE}i\max} \end{cases} \end{cases} \tag{18}$$

RO model is another model that focuses on worst effects on system stability caused by uncertainty. It is necessary to explain similarities and differences between the proposed model and traditional RO model [25], [26]. Compared with RO model, the common point is modeling of uncertain fluctuations of RES outputs is the same. They do not require accurate modeling of probability distribution of uncertain parameters in the optimization model [27]. Difference is proposed model describes uncertainty of SVSM as an interval, not only for worst case of RES outputs, but also for best case of RES outputs. The obtained upper bound and its corresponding RES output values can supply some useful information about how to increase the system SVSM by adjusting RES outputs. By comparison, RO model is more suitable for optimal control problem of power system, interval optimization model is more suitable for analysis problem of power system. Calculation of SVSM is attributed to the analysis problem of power system, hence interval optimization model is used in this paper.

III. Convex Solution Method of the Worst-case SVSM Interval Identification Model

A. Transformation of Three-level Optimization Model

In TLOPF model (17), outer and middle level optimization problems are both max problems, which can be transformed into a single-level model. Thus, (17) can be transformed into a bi-level max–min optimization model, expressed in (19).

$$\begin{cases} \max\limits_{\boldsymbol{P}_{\mathrm{RE}},\phi} \min\limits_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \\ \text{s.t.} \begin{cases} \text{(2b), (4)–(6), (7a), (7b), (8a), (8b),} \\ \quad \text{(9)–(11), (13)–(16)} \\ P_{\mathrm{RE}i\min} \le P_{\mathrm{RE}i} \le P_{\mathrm{RE}i\max} \end{cases} \end{cases} \tag{19}$$

In TLOPF model (18), outer level and middle level optimization problems are a min problem and a max problem, respectively, so they cannot be directly merged. To solve this problem, order of outer level and middle level optimization problems can be exchanged. Since RES-connected buses are generally not directly connected to loads in actual engineering, there are no terms of load growth $\lambda P_{\mathrm{L}i0}\phi_i$ and $\lambda Q_{\mathrm{L}i0}\phi_i$ in power balance equations of these buses, whereas there are no terms of $P_{\mathrm{RE}i}$ and $Q_{\mathrm{RE}i}$ in power balance equations of load buses. Therefore, terms relative to $\boldsymbol{P}_{\mathrm{RE}}$ and ϕ do not exist in the same constraint of the proposed OPF model, that is, decision variables $\boldsymbol{P}_{\mathrm{RE}}$ and ϕ are independent of each other, and order exchange does not affect optimization results, as shown in (20). It can be seen after order exchange of outer level and middle level optimization problems, inner level and exchanged middle level optimization problems are both min problems, hence they can be directly merged. Thus, TLOPF model (18) can be transformed into a bi-level max–min optimization model, expressed in (21).

$$\min_{\boldsymbol{P}_{\mathrm{RE}}} \max_{\phi} \min_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \Leftrightarrow \max_{\phi} \min_{\boldsymbol{P}_{\mathrm{RE}}} \min_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \tag{20}$$

$$\begin{cases} \max\limits_{\phi} \min\limits_{\boldsymbol{P}_{\mathrm{RE}},\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \\ \text{s.t.} \begin{cases} \text{(2b), (4)–(6), (7a), (7b), (8a), (8b),} \\ \quad \text{(9)–(11), (13)–(16)} \\ P_{\mathrm{RE}i\min} \le P_{\mathrm{RE}i} \le P_{\mathrm{RE}i\max} \end{cases} \end{cases} \tag{21}$$

Difference between the two bi-level max–min optimization models expressed in (19) and (21) is RES output $\boldsymbol{P}_{\mathrm{RE}}$ is an outer-level variable of the model in (19), but an inner-level variable in (21). Therefore, for inner level model of (19), $\boldsymbol{P}_{\mathrm{RE}}$ is constant.

B. Convex Solution Method of the Bi-Level Optimization Model

1) Duality of Convex Programming Problems

Assume object function $f(\boldsymbol{x})$ and constraints $\boldsymbol{h}(\boldsymbol{x})$ are convex in (22). According to [28], (22) is a convex programming problem, and nonlinear programming problem (23) is dual form of original optimization (22).

$$\begin{cases} \min\limits_{\boldsymbol{x}} f(\boldsymbol{x}) \\ \text{s.t. } \boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0} \end{cases} \tag{22}$$

$$\begin{cases} \max\limits_{\boldsymbol{x},\boldsymbol{v}} g(\boldsymbol{x},\boldsymbol{v}) = f(\boldsymbol{x}) + \boldsymbol{v}^{\mathrm{T}}\boldsymbol{h}(\boldsymbol{x}) \\ \text{s.t. } \boldsymbol{v}^{\mathrm{T}}\dfrac{\mathrm{d}\boldsymbol{h}}{\mathrm{d}\boldsymbol{x}} + \dfrac{\mathrm{d}f}{\mathrm{d}\boldsymbol{x}} = 0,\ v_i \ge 0 \end{cases} \tag{23}$$

where $\boldsymbol{v} = (v_1, v_2, \cdots, v_N)^{\mathrm{T}}$, v_i is dual variable, N is number of constraints, and $\boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0}$.

As long as inner level models in (19) and (21) are transformed into their dual max models, bi-level programming problems (19) and (21) become max–max problems, and they can be transformed into single-level max programming problems. However, some constraints of inner level problems in (19) and (21) are non-convex, hence different convex relaxation methods are applied in (19) and (21).

For inner level models in (19) and (21), variables of outer models can be considered as constants, that is, $\boldsymbol{\phi}$ is constant for inner level models in (19) and (21). Therefore, objective functions of inner level min models in (19) and (21) are linear and do not require convex relaxation. Only nonlinear constraints in the two bi-level programming models require convex relaxation.

2) Convex Relaxation of Inner Level Optimization Model

a) Convex Relaxation of Power Flow Equations: According to [28], SOCP relaxation can be used for power flow equations by introducing the following variables:

$$R_{ij} = V_i V_j \cos\theta_{ij},\ I_{ij} = V_i V_j \sin\theta_{ij},\ U_i = V_i^2. \tag{24}$$

Meanwhile, first-order Taylor series expansion is used to approximate the equation stating the sum of the phase angle difference of each branch in an independent loop circuit is equal to 0. Then (4)–(6) are relaxed to linear equality and SOC inequality constraints, as follows:

$$\begin{cases} P_{\mathrm{G}i} + P_{\mathrm{RE}i} - (P_{\mathrm{L}i0} + \lambda P_{\mathrm{L}i0}\phi_i) - U_i G_{ii} \\ \quad - \sum_{j=1}^{n}(R_{ij}G_{ij} + I_{ij}B_{ij}) = 0 \\ Q_{\mathrm{G}i} + Q_{\mathrm{RE}i} - (Q_{\mathrm{L}i0} + \lambda Q_{\mathrm{L}i0}\phi_i) + U_i B_{ii} \\ \quad - \sum_{j=1}^{n}(I_{ij}G_{ij} - R_{ij}B_{ij}) = 0 \\ -U_i G_{ii} - \sum_{j=1}^{n}(R_{ij}G_{ij} + I_{ij}B_{ij}) \pm K_{\mathrm{p}i}U_{\mathrm{d}i}I_{\mathrm{d}i} = 0 \\ U_i B_{ii} - \sum_{j=1}^{n}(I_{ij}G_{ij} - R_{ij}B_{ij}) - K_{\mathrm{p}i}U_{\mathrm{d}i}I_{\mathrm{d}i}\tan\phi_i = 0 \\ -U_i G_{ii} - \sum_{j=1}^{n}(R_{ij}G_{ij} + I_{ij}B_{ij}) - P_{\mathrm{v}i} = 0 \\ U_i B_{ii} - \sum_{j=1}^{n}(I_{ij}G_{ij} - R_{ij}B_{ij}) - Q_{\mathrm{v}i} = 0 \\ \|2R_{ij};\ 2I_{ij};\ U_i - U_j\|_2 \le U_i + U_j \\ R_{ij} = R_{ji}, I_{ij} = -I_{ji} \\ \sum_{ij\in C_k}\left[\tan^{-1}\dfrac{I_{ij}^0}{R_{ij}^0} + \dfrac{1}{(R_{ij}^0)^2 + (I_{ij}^0)^2}(R_{ij}^0 I_{ij} - I_{ij}^0 R_{ij})\right] \\ \quad = 0 \end{cases} \tag{25}$$

b) Convex Relaxation of Complementary Equations: According to convex relaxation for power flow equations, the square form of $V_{\mathrm{G}i}$, i.e., $U_{\mathrm{G}i}$, replaces $V_{\mathrm{G}i}$ in the model. Hence, complementary equations of $Q_{\mathrm{G}i}$ and $V_{\mathrm{G}i}$ (11a)–(11e) are rewritten as (26a)–(26e).

$$Q_{\mathrm{G}i\min} \le Q_{\mathrm{G}i} \le Q_{\mathrm{G}i\max} \tag{26a}$$

$$U_{\mathrm{G}i} = V_{\mathrm{G}i\mathrm{ref}}^2 + U_{\mathrm{a}i} - U_{\mathrm{b}i} \tag{26b}$$

$$(Q_{\mathrm{G}i} - Q_{\mathrm{G}i\min})U_{\mathrm{a}i} = 0 \tag{26c}$$

$$(Q_{\mathrm{G}i\max} - Q_{\mathrm{G}i})U_{\mathrm{b}i} = 0 \tag{26d}$$

$$U_{\mathrm{a}i} \ge 0, U_{\mathrm{b}i} \ge 0 \tag{26e}$$

Equations (26c) and (26d) are non-convex equalities, which can be transformed into convex constraints by convex hull relaxation method [29]. For the square term of a variable x, it can be expressed by auxiliary variable z_1, and T hull relaxation expression is shown in (27). For product term of two variables x and y, it can be expressed by auxiliary variable z_2, and M hull relaxation expression is shown in (28). Equations (27) and (28) can be simplified to the form shown in (29) [30].

$$\begin{cases} x^2 - z_1 \le 0 \\ z_1 - (x_{\max} + x_{\min})x + x_{\max}x_{\min} \le 0 \end{cases} \tag{27}$$

$$\begin{cases} x_{\min}y + y_{\min}x - x_{\min}y_{\min} - z_2 \le 0 \\ x_{\max}y + y_{\max}x - x_{\max}y_{\max} - z_2 \le 0 \\ z_2 - x_{\min}y - y_{\max}x + x_{\min}y_{\max} \le 0 \\ z_2 - x_{\max}y - y_{\min}x + x_{\max}y_{\min} \le 0 \end{cases} \tag{28}$$

$$z_1 = \langle x^2\rangle^{\mathrm{T}},\ z_2 = \langle xy\rangle^{\mathrm{M}} \tag{29}$$

Let $q_{\mathrm{a}i} = Q_{\mathrm{G}i}U_{\mathrm{a}i}$ and $q_{\mathrm{b}i} = Q_{\mathrm{G}i}U_{\mathrm{b}i}$. Then (26a) and (26c)–(26e) are transformed into the following linear forms:

$$\begin{cases} q_{\mathrm{a}i} - Q_{\mathrm{G}i\min}U_{\mathrm{a}i} = 0,\ q_{\mathrm{a}i} = \langle Q_{\mathrm{G}i\min}U_{\mathrm{a}i}\rangle^{\mathrm{M}} \\ Q_{\mathrm{G}i\max}U_{\mathrm{b}i} - q_{\mathrm{b}i} = 0,\ q_{\mathrm{b}i} = \langle Q_{\mathrm{G}i\max}U_{\mathrm{b}i}\rangle^{\mathrm{M}} \end{cases} \tag{30}$$

c) Convex Relaxation of LCC-HVDC Line Operation Characteristic Constraints: Assume ratio $K_{\mathrm{d}i}$ does not change with load growth. There is a nonlinear term $V_i\cos\xi_i$ in (7a). By regarding $\cos\xi_i$ as a whole continuous variable and letting $w_i = V_i\cos\xi_i$, then (7a) can be transformed into the following linear constraint by convex hull relaxation method:

$$\begin{cases} U_{\mathrm{d}i} = 3\sqrt{2}/\pi \cdot K_{\mathrm{d}i}w_i - X_{\mathrm{c}i}I_{\mathrm{d}i} \\ w_i = \langle V_i\cos\xi_i\rangle^{\mathrm{M}} \end{cases} \tag{31}$$

Equation (7b) also has nonlinear terms V_i^2 and $\tan^2\phi_i$. Since $U_i = V_i^2$ in the convex process of AC power flow equation, this nonconvex quadratic equation needs to be added in the OPF model, and it can be transformed into the following convex inequality constraint:

$$U_i = \langle V_i^2\rangle^{\mathrm{T}} \tag{32}$$

Regard $\tan\varphi_i$ as a whole continuous variable, and let $U_{\mathrm{t}i} = U_{\mathrm{d}i}\tan\varphi_i$, $T_i = U_{\mathrm{t}i}^2$, and $D_i = U_{\mathrm{d}i}^2$. Similarly, these nonconvex quadratic equations are transformed into corresponding convex inequality constraints as follows:

$$U_{\mathrm{t}i} = \langle U_{\mathrm{d}i}\tan\phi_i\rangle^{\mathrm{M}},\ T_i = \langle U_{\mathrm{t}i}^2\rangle^{\mathrm{T}},\ D_i = \langle U_{\mathrm{d}i}^2\rangle^{\mathrm{T}} \tag{33}$$

Thus, (7b) can be transformed into a linear equality constraint:

$$T_i + D_i - \left(3\sqrt{2}/\pi \cdot K_{\mathrm{d}i}\right)^2 U_i = 0 \tag{34}$$

In addition, (13a) describing control mode switching of the LCC-HVDC inverter side is relaxed by convex hull relaxation method. Let $s_i = V_{\mathrm{d}i}\cos\xi_{\mathrm{I}i}$, and then (13a) is transformed into the following linear constraints:

$$\begin{cases} \cos\xi_{\mathrm{I}i\,\min}V_{\mathrm{d}i} - s_i = 0 \\ s_i = \langle V_{\mathrm{d}i}\cos\xi_{\mathrm{I}i}\rangle^{\mathrm{M}} \end{cases} \tag{35}$$

d) Convex Relaxation of VSC-HVDC Line Operation Characteristic Constraints: Equation (8) can also be transformed into convex constraints by convex hull relaxation method. Since phase angle difference δ_i between AC bus voltage and VSC voltage at AC side is a small value, sin

$\delta_i \approx \delta_i$ and $\cos\delta_i \approx 1$. Let $O_i = M_i^2$. Then it can be transformed into the following convex inequality constraint:

$$O_i = \langle M_i^2 \rangle^{\mathrm{T}} \tag{36}$$

Let $L_{\mathrm{v}i} = M_i V_i$ and $K_{\mathrm{v}i} = M_i V_i \delta_i = L_{\mathrm{v}i}\delta_i$. Then (8) can be transformed into linear constraints as follows:

$$\begin{cases} P_{\mathrm{vd}i} + \dfrac{\mu_i}{\sqrt{2}} U_{\mathrm{d}i} Y_i (L_{\mathrm{v}i}\cos\alpha_i - K_{\mathrm{v}i}\sin\alpha_i) - U_i Y_i \cos\alpha_i = 0 \\ Q_{\mathrm{vd}i} + \dfrac{\mu_i}{\sqrt{2}} U_{\mathrm{d}i} Y_i (K_{\mathrm{v}i}\cos\alpha_i + L_{\mathrm{v}i}\sin\alpha_i) - U_i Y_i \sin\alpha_i = 0 \\ U_{\mathrm{d}i} I_{\mathrm{d}i} - \dfrac{\mu_i}{\sqrt{2}} \cdot U_{\mathrm{d}i} Y_i (L_{\mathrm{v}i}\cos\alpha_i + K_{\mathrm{v}i}\sin\alpha_i) \\ \quad + \left(\dfrac{\mu_i}{\sqrt{2}} \cdot U_{\mathrm{d}i}\right)^2 O_i Y_i \cos\alpha_i = 0 \\ L_{\mathrm{v}i} = \langle M_i V_i \rangle^{\mathrm{M}}, K_{\mathrm{v}i} = \langle L_{\mathrm{v}i}\delta_i \rangle^{\mathrm{M}} \end{cases} \tag{37}$$

Equation (14) describing control mode switching of VSC-HVDC inverter side can also be relaxed by convex hull relaxation method. Let $t_i = Q_{\mathrm{e}i} V_i$. Then it can be transformed into the following convex constraints:

$$\begin{cases} V_{i\min} Q_{\mathrm{e}i} - t_i = 0 \\ t_i = \langle V_i Q_{\mathrm{e}i} \rangle^{\mathrm{M}} \end{cases} \tag{38}$$

e) Inner Level Model After Convex Relaxation: After the abovementioned convex relaxation transformation, inner level models in (19) and (21) are transformed into convex programming models, expressed in (39) and (40).

$$\begin{cases} \min\limits_{\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \\ \text{s.t.} \begin{cases} \text{(2b), (10), (15), (16), (25), (26b), (30)} & \\ \text{(13b)–(13d), (31)–(35)} & i \in N_{\mathrm{LCC}} \\ \text{(14b), (14c), (32), (36)–(38)} & i \in N_{\mathrm{VSC}} \end{cases} \end{cases} \tag{39}$$

$$\begin{cases} \min\limits_{\boldsymbol{P}_{\mathrm{RE}},\boldsymbol{x}} -\sum_{i=1}^{n} \lambda P_{\mathrm{L}i0}\phi_i \\ \text{s.t.} \begin{cases} \text{(2b), (10), (15), (16), (25), (26b), (30)} & \\ \text{(13b)–(13d), (31)–(35)} & i \in N_{\mathrm{LCC}} \\ \text{(14b), (14c), (32), (36)–(38)} & i \in N_{\mathrm{VSC}} \\ P_{\mathrm{RE}i\min} \le P_{\mathrm{RE}i} \le P_{\mathrm{RE}i\max} & \end{cases} \end{cases} \tag{40}$$

3) Transformed Single-Level Optimization Model

For inner level convex programming models (39) and (40), after convex relaxation, dual programming modes can be constructed according to (23). Thus, the two bi-level max–min problems (19) and (21) are both transformed into max–max bi-level problems, which can be directly merged into single-level programming problems (41) and (42), respectively. For (40), $\boldsymbol{P}_{\mathrm{RE}}$ is a variable but not a constant, and change of $\boldsymbol{P}_{\mathrm{RE}}$ should be considered when constructing dual programming problem of (40). Thus, objective function in dual programming model (42) has one more term on $\boldsymbol{P}_{\mathrm{RE}}$ than in (41), and constraints in (42) have some derivative constraints on $\boldsymbol{P}_{\mathrm{RE}}$.

$$\begin{cases} \max\limits_{\boldsymbol{x},\boldsymbol{v},\boldsymbol{P}_{\mathrm{RE}},\boldsymbol{\phi}} g(\boldsymbol{x},\boldsymbol{v}) \\ \text{s.t. } \boldsymbol{v}^{\mathrm{T}} \dfrac{\mathrm{d}\boldsymbol{h}}{\mathrm{d}\boldsymbol{x}} + \dfrac{\mathrm{d}f}{\mathrm{d}\boldsymbol{x}} = 0, \ v_i \ge 0 \end{cases} \tag{41}$$

$$\begin{cases} \max\limits_{\boldsymbol{x},\boldsymbol{v},\boldsymbol{P}_{\mathrm{RE}},\boldsymbol{\phi}} g(\boldsymbol{x},\boldsymbol{v}) + \boldsymbol{v}_{\mathrm{RE}}^{\mathrm{T}} \boldsymbol{h}_{\mathrm{RE}}(\boldsymbol{P}_{\mathrm{RE}}) \\ \text{s.t.} \begin{cases} \boldsymbol{v}^{\mathrm{T}} \dfrac{\mathrm{d}\boldsymbol{h}}{\mathrm{d}\boldsymbol{x}} + \dfrac{\mathrm{d}f}{\mathrm{d}\boldsymbol{x}} = 0, & v_i \ge 0 \\ \boldsymbol{v}_{\mathrm{RE}}^{\mathrm{T}} (\mathrm{d}\boldsymbol{h}_{\mathrm{RE}}/\mathrm{d}\boldsymbol{P}_{\mathrm{RE}}) = 0, & v_{\mathrm{RE}i} \ge 0 \end{cases} \end{cases} \tag{42}$$

where $\boldsymbol{h}_{\mathrm{RE}}$ represents constraints containing $\boldsymbol{P}_{\mathrm{RE}}$ in the original model (40), i.e., active power flow equations of the RES buses, and $\boldsymbol{v}_{\mathrm{RE}}$ represents dual variables corresponding to these constraints. These two single-level optimization problems are non-linear programming problems, which can be solved by solver CONOPT in commercial optimization software GAMS [31].

IV. Case Studies

A. Modified IEEE-39 Bus AC/DC System

Modified IEEE-39-bus AC/DC power system is taken as an example. AC line 14–4 is replaced with a two-terminal LCC-HVDC line, AC lines 11–6 and 11–12 are replaced with a three-terminal VSC-HVDC system, and an RES is connected at bus 9, as shown in Fig. 2. Upper and lower limits of AC bus voltage are 1.1 and 0.9. For the two-terminal LCC-HVDC transmission line, $R_{\mathrm{dc}} = 0.02$, $X_{\mathrm{c}} = 0.1$; on rectifier side, $K_{\mathrm{d}4} = 1.1$, and on inverter side, $K_{\mathrm{d}14} = 1.05$; upper and lower limits of θ_i are 15° and 8°, respectively; and upper and lower limits of $\tan\varphi_i$ are 0.6 and 0.4, respectively. For three-terminal VSC-HVDC transmission system, resistance of each line is $R_{\mathrm{dc}} = 0.01$, and equivalent admittance and impedance angle of the VSC-HVDC converter station are $Y_i = 6.661$ and $\alpha_i = 1.53$. Upper and lower limits of M_i are 1 and 0.8, respectively; upper and lower limits of δ_i are 5° and 0°, respectively. For LCC/VSC-HVDC stations, control modes in the above subsection 2.1 are adopted. Combined DC line operation constraints (7c) and (8d), all DC variables can be calculated, and they are shown in Table I. Reactive powers $Q_{\mathrm{vd}6}$, $Q_{\mathrm{vd}11}$, and $Q_{\mathrm{vd}12}$ are −0.80, −0.13, and −0.30, respectively. The

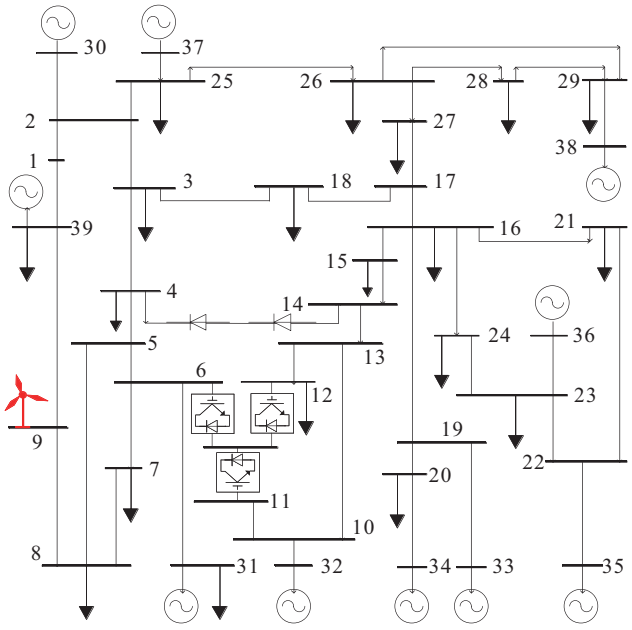


Fig. 2. Modified IEEE 39-bus AC/DC system.

TABLE I
DC VARIABLE VALUES OF HVDC TRANSMISSION LINES

Bus	$U_{\mathrm{d}i}$	$I_{\mathrm{d}i}$
4	1.2394	0.53
14	1.2500	0.53
6	1.9974	−0.05
11	2.0000	0.08
12	1.9978	−0.03

computer used for the case study is a DELL OptiPlex 3010 workstation with a 3.50 GHz Intel Xeon processor E3-1270 and 32 GB RAM.

1) Comparative Results of Different LGD Models

In order to obtain reasonable worst-case LGD within the possible load growth in actual situations, in the hypercone model, uncertain deviation degree of each load bus β_i is set to a reasonable value between 0 and 1. When P_{RE9} takes central value of the given fluctuation interval 1000 MW, Table II shows comparative results obtained by hypercone model and conventional model without considering constraint of deviation degree of actual LGD from initial LGD, that is $\beta_i = 1$ and $\phi > 0$. It can be seen with increase of β_i, "thickness" of the hypercone increases, value of worst-case SVSM reduces. In the hypercone model, worst case for the model obtained in this table is when $\beta_i = 0.9$, worst-case SVSM is 272.77 MW. While in conventional model, worst case LGD reach maximum deviation degree from initial LGD, worst-case SVSM obtains minimal value 120.74 MW. In this LGD, only the load of bus 4 increases, the value of ϕ_4 is not 0, and other load buses have no load increase. Because bus 4 belongs to both heavy load bus and LCC-HVDC inverter station's AC bus in the system, load growth of bus 4 is easier to reach SVS limit point, and this LGD is considered to be worst case under conventional model. Due to the fact the actual LGD often has a certain deviation degree from initial LGD, the hypercone model can better reflect actual situation compared with conventional model.

TABLE II
COMPARISON OF THE WORST-CASE SVSM BY DIFFERENT LGD MODELS

LGD model		Worst-case SVSM (MW)	CPU time (s)
Hypercone Model	$\beta_i = 0.1$	858.96	0.29
	$\beta_i = 0.3$	654.36	0.36
	$\beta_i = 0.5$	499.05	0.59
	$\beta_i = 0.7$	374.61	0.47
	$\beta_i = 0.9$	272.77	0.51
Conventional Model	$\beta_i = 1$ and $\phi > 0$	120.74	0.18

2) Analysis of the Results by the Proposed Method

Assume RES adopts control mode of constant power factor, and power factor is 1.0. Uncertain deviation degree of the hypercone model of each load bus is $\beta_i = 0.5$. Given different fluctuation intervals of RES output, proposed TLOPF calculation method, traditional Monte Carlo sampling (MCS) method, and Latin hypercube sampling (LHS) method, and affine arithmetic (AA) method are used to identify worst-case SVSM interval of the system. In these two sampling methods, MCS method generates 1000 uniformly distributed RES output samples in a given RES output interval. LHS method divides value space of the random variable P_{RE9} into 300 subspaces, and independent sampling is executed in each subspace. For each sample of RES output, worst-case SVSM of AC/DC power system is obtained by solving deterministic bi-level OPF model in Section 2.1. Maximum and minimum of worst-case SVSMs corresponding to all samples are obtained to form worst-case SVSM interval. In AA method [14], [16], the proposed bi-level OPF model in Subsection 2.1 is used to calculate worst-case SVSM and corresponding SVS limit point with $P_{\mathrm{RE9}} = 1000$ MW, and sensitivity of worst-case SVSM with respect to P_{RE9} is calculated at this SVS limit point. Calculation results obtained by the four tested methods are shown in Table III.

TABLE III
COMPARISON OF THE WORST-CASE SVSM INTERVALS OBTAINED BY FOUR METHODS

P_{RE9} interval (MW)	Method	SVSM interval (MW)	P_{RE9} of the lower bound (MW)	P_{RE9} of the upper bound (MW)	CPU time (s)
[540, 1460]	TLOPF	[323.9, 503.8]	1460.0	905.0	1.57
	MCS	[324.2, 503.7]	1459.1	899.4	380.47
	LHS	[324.0, 503.7]	1459.6	902.3	96.48
	AA	[321.6, 507.4]	–	–	2.12
[600, 1400]	TLOPF	[362.9, 503.8]	1400.0	905.0	1.67
	MCS	[363.2, 503.7]	1399.2	901.8	378.00
	LHS	[363.0, 503.8]	1399.8	903.4	99.12
	AA	[360.8, 508.2]	–	–	2.04
[650, 1350]	TLOPF	[391.4, 503.8]	1350.0	905.0	1.61
	MCS	[391.7, 503.7]	1349.1	901.1	377.23
	LHS	[391.5, 503.8]	1349.4	904.6	99.94
	AA	[389.7, 507.6]	–	–	2.06

It can be seen that with increase of uncertain fluctuation range of P_{RE9}, range of the worst-case SVSM gradually increases. When $P_{\mathrm{RE9}} = 905$ MW, worst-case SVSM has maximum value. Minimum value of worst-case SVSM generally corresponds to maximum value of given RES output interval. In addition, it can be observed the worst-case SVSM intervals obtained by the proposed TLOPF method have few errors compared with MCS and LHS methods. Absolute errors of the two bounds of the obtained SVSM interval are both less than 0.003, which indicates the worst-case SVSM interval calculated by the proposed TLOPF method has high computational accuracy. In addition, CPU time of the proposed TLOPF model is much lower than CPU times of MCS and LHS methods, as TLOPF method only takes about 1.6 s, while LHS method takes nearly 100 s, and MCS method takes more than 300 s. CPU time of the AA method is close to that of the proposed TLOPF method, which is about 2 s. However, obtained worst-case SVSM interval of AA method has obviously bigger errors than of the proposed TLOPF method with results of two sampling methods as the benchmark, and its fluctuation range is obviously bigger than the other three methods. Moreover, AA method cannot directly obtain RES outputs corresponding to the upper and lower bounds of worst-case SVSM interval. Therefore, the proposed TLOPF method has much higher computational efficiency for calculating worst-case SVSM interval of AC/DC hybrid grid than MCS, LHS methods and has much higher computational accuracy than AA method.

The worst-case LGD can be obtained by the ϕ_i value of each load bus corresponding to optimal solution of the proposed TLOPF model. Taking lower bound of the worst-case SVSM interval when RES output is $P_{\mathrm{RE9}} = 1460$ MW as an example, obtained worst-case LGD is shown in Fig. 3. It can be seen in the obtained worst-case LGD, load growth deviation coefficient ϕ_i takes maximum value corresponding to heavy load buses, such as buses 4, 7, and 8. Since load growths of these buses are more likely to result in substantial decreases in their bus voltages, and these buses are SVS weak buses of the system [32], taking maximum value of their ϕ_i will make the system more easily reach SVS limit point, resulting in voltage instability.

Variation voltage and extinction angle of the LCC-HVDC inverter station are plotted against load growth in Fig. 4. When RES output is $P_{\mathrm{RE9}} = 905$ MW, load growth along worst-case LGD causes control mode switching of LCC-HVDC inverter station. Control mode switches from DC voltage to constant extinction angle when extinction angle of the LCC-HVDC inverter station reaches minimum secure limit value $\xi_{\mathrm{I}i\min}$ with load growth. Before control mode switching, U_{d4} is constant and ξ_4 decreases with load growth. After control mode switching, ξ_4 is constant and U_{d4} decreases with load growth. AC bus voltage V_4 always decreases with load growth. This shows the proposed worst-case SVSM calculation model of the AC/DC power system can effectively consider control mode switching of DC converter station with load growth.

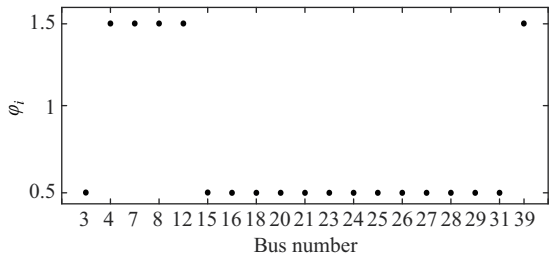


Fig. 3. Obtained worst-case LGD.

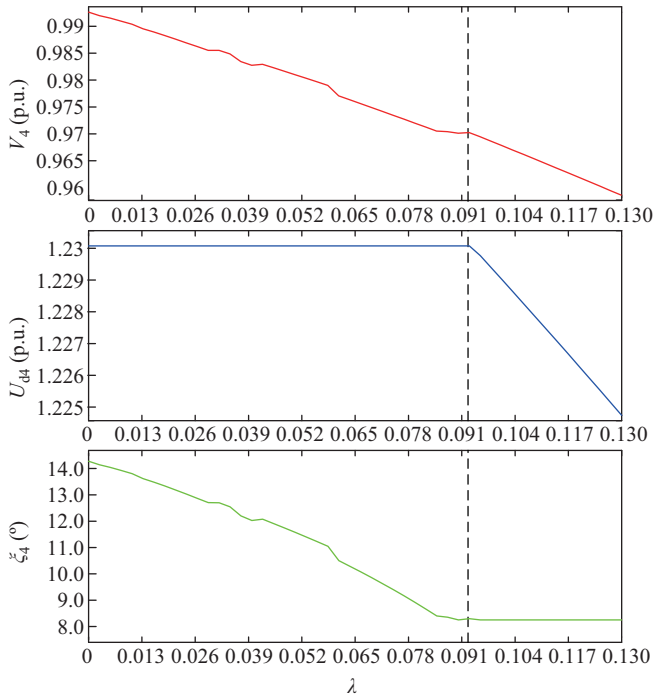


Fig. 4. Changes of the LCC station variables with load growth.

To reflect system operation risk caused by calculation results without considering uncertainty of LGD, LGD is fixed as the initial LGD, i.e., $\phi_i = 1.0$ for each load bus, and identification models are bi-level OPF models, as in [17]. Given RES output fluctuation interval is the same as Table III. BLOPF, MCS, LHS and AA methods are used to calculate SVSM interval of the system. For each RES output sample in sampling methods, corresponding SVSM of the AC/DC power system under initial LGD is obtained by solving single-level OPF model, i.e., inner level model of bi-level OPF model in Section 2.1. Maximum and minimum SVSM values of all samples are obtained to form SVSM interval. Results are shown in Table IV.

TABLE IV
COMPARISON OF OBTAINED SVSM INTERVALS OF INITIAL LGD BY FOUR METHODS

P_{RE9} interval (MW)	Method	SVSM interval (MW)	P_{RE9} of the lower bound (MW)	P_{RE9} of the upper bound (MW)	CPU time (s)
[540, 1460]	BLOPF	[648.6, 980.3]	1460.0	927.2	1.85
	MCS	[648.7, 980.2]	1459.3	921.9	431.12
	LHS	[648.7, 980.3]	1459.4	925.4	98.20
	AA	[644.2, 982.4]	–	–	2.08
[600, 1400]	BLOPF	[724.1, 980.3]	1400.0	927.2	1.89
	MCS	[724.3, 980.2]	1399.1	923.1	388.59
	LHS	[724.2, 980.3]	1399.5	926.1	99.56
	AA	[721.5, 983.1]	–	–	2.18
[650, 1350]	BLOPF	[778.6, 980.3]	1350.0	927.2	1.92
	MCS	[778.7, 980.3]	1349.1	925.8	389.85
	LHS	[778.6, 980.3]	1349.8	926.7	99.91
	AA	[777.2, 982.8]	–	–	2.07

Comparing SVSM interval results in Tables III and IV, it can be seen the SVSM interval under worst-case LGD is significantly smaller than under initial LGD. For example, when P_{RE9} is [540, 1460] MW, lower bound of SVSM interval decreases from [648.6, 980.3] MW under initial LGD to [323.9, 503.8] MW under worst-case LGD; the load growth under worst-case LGD is more likely to reach SVS limit point of the system. Therefore, if only considering SVSMs under initial LGD, actual system operation may face risk of static voltage instability due to uncertainty of LGD.

3) Comparative Results of Different Integration Rates of Renewable Energy

A comparison of results under different integration rates of renewable energy is shown in Table V. RES output fluctuation range is set as ±25% of given central value of integration rate, and uncertain LGD is set as $\beta_i = 0.5$. In Table V, when integration rate of renewable energy is less than 15%, with increase in integration rate, overall value of worst-case SVSM interval increases, and fluctuation range also increases. When integration rate is equal to 15%, since RES output interval contains $P_{\mathrm{RE9}} = 905.0$ MW corresponding to maximum point of SVSM, upper bound of worst-case SVSM achieve maximum value. When integration rate is 20%, since RES

TABLE V
COMPARISON OF WORST-CASE SVSM INTERVALS UNDER DIFFERENT INTEGRATION RATES OF RENEWABLE ENERGY

Integration rate	$P_{\mathrm{RE}9}$ interval (MW)	SVSM interval (MW)	$P_{\mathrm{RE}9}$ of the lower bound (MW)	$P_{\mathrm{RE}9}$ of the upper bound (MW)	CPU time (s)
5%	[231, 384]	[294.1, 375.7]	231	384	1.545
10%	[461, 768]	[409.7, 494.4]	461	768	1.625
15%	[692, 1153]	[470.5, 503.8]	1153	905	1.642
20%	[923, 1538]	[264.6, 503.6]	1538	923	1.597

output interval is greater than 905.0 MW, worst-case SVSM decreases with increase of $P_{\mathrm{RE}9}$.

B. Comparative Results of Different Setting of RESs

Given RES output fluctuation interval of [540 MW, 1460 MW], a comparison of obtained worst-case SVSM intervals under different power factors of RES is shown in Table VI. With decrease in power factor of RES, overall value of worst-case SVSM interval increases, and RES output value corresponding to upper bound increases. This is because as power factor of RES decreases, RES provides more reactive power to AC/DC power system, which can improve SVS of the whole system.

The above analyses of RES output fluctuation are based on RES connected to bus 9 as shown in Fig. 2. Obviously, the proposed model can also analyze worst-case SVSM interval for different RES connected locations and different number of RESs. Note the RES connected at bus 9 is RES 1, and given active power fluctuation interval is [700 MW, 1300 MW]. Two RES respectively connected at buses 1 and 18 are added and named as RES 2 and RES 3, their given active power fluctuation intervals are the same as RES 1. It is assumed there is no correlation between active power outputs of the three RESs. Comparison of obtained worst-case SVSM intervals under different RES connected locations and different number of RESs is shown in Table VII and VIII. Obtained worst-case SVSM intervals and corresponding RES outputs vary with different RES connected locations and different number of RESs. In Table VII, different RES connected locations have

TABLE VI
COMPARISON OF OBTAINED WORST-CASE SVSM INTERVALS UNDER DIFFERENT POWER FACTORS OF RESS

Power factor of RES	SVSM interval (MW)	$P_{\mathrm{RE}9}$ of the lower bound (MW)	$P_{\mathrm{RE}9}$ of the upper bound (MW)	CPU time (s)
1.00	[323.9, 503.8]	1460.0	905.0	1.570
0.95	[623.4, 960.9]	540.0	1431.7	1.511
0.90	[705.4, 1217.0]	540.0	1460.0	1.622
0.85	[764.3, 1425.1]	540.0	1460.0	1.634

TABLE VII
COMPARISON OF THE OBTAINED WORST-CASE SVSM INTERVALS UNDER DIFFERENT RES CONNECTED LOCATIONS

RES Connection	SVSM interval (MW)	$P_{\mathrm{RE}9}$ of the lower bound (MW)	$P_{\mathrm{RE}9}$ of the upper bound (MW)
1	[416.3, 503.8]	905.0	1300.0
2	[305.4, 398.1]	1262.2	700.0
3	[106.0, 150.6]	700.0	1300.0

TABLE VIII
COMPARISON OF THE OBTAINED WORST-CASE SVSM INTERVALS UNDER DIFFERENT NUMBER OF RESS

RES Connection	SVSM interval (MW)	$P_{\mathrm{RE}9}$ of the lower bound (MW)	$P_{\mathrm{RE}9}$ of the upper bound (MW)
1	[416.3, 503.8]	905.0	1300.0
1, 2	[349.2, 494.9]	(700.0, 700.0)	(1300.0, 1300.0)
1, 2, 3	[127.5, 455.1]	(777.8, 700.0, 1300.0)	(1300.0, 1300.0, 700.0)

different effects on the SVS. Overall value of obtained worst-case SVSM intervals in RES 1 and RES 2 are bigger than in RES 3. It is because RES 1 and RES 2 are near generator bus 39. Adjustment of generator output can balance uncertain fluctuation of RES outputs. There is no generator near bus 18 where uncertain fluctuation of RES outputs can directly affect load growth, hence the obtained worst-case SVSM interval when RES is connected at bus 18 is smaller. In Table VIII, as number of RESs increases, total fluctuation range of RES outputs also increases, resulting in increase of fluctuation range of obtained worst-case SVSM intervals.

C. Actual AC/DC Power Grid in China

The second case is AC/DC power system of China Southern Power Grid (CSPG) in 2020, which includes 8733 buses, 11,240 branches, and 614 generators. The main wiring diagram of CSPG is shown in Fig. 5, which includes four provincial power grids. It also includes 10 LCC-HVDC transmission lines and 1 three-terminal VSC-HVDC transmission line [33].

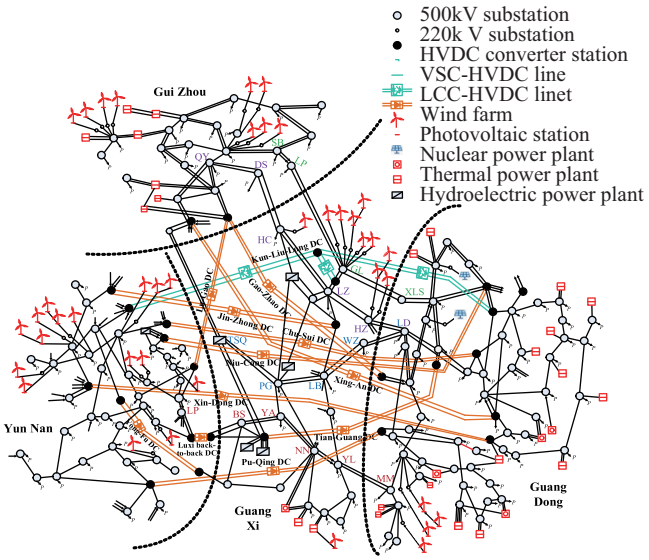


Fig. 5. Main wiring diagram of the actual AC/DC power grid in China.

There are 42 RESs in the grid, including 40 wind farms (WFs) and 2 photovoltaic (PV) power stations. Location distribution and parameters of RESs are shown in Table IX, where central values of RES output fluctuation intervals are active power output values of RES in operation data of CSPG. Assume the power factor of each RES is 1.0. Given uncertain RES output fluctuation intervals are $[k_{\min}, k_{\max}] * P_{\mathrm{RE}i0}$, where $P_{\mathrm{RE}i0}$ is central value of interval and is presented in last column in Table IX, and $k_{\min}$ and $k_{\max}$ are minimum and

TABLE IX
DISTRIBUTION AND PARAMETERS OF RESS

No.	Area	Type	Bus Name	Rated Capacity (MW)	Central Value of $\boldsymbol{P}_{\mathrm{RE}}$ Interval (MW)
1		WF	XINYU22	24	20.0
2		WF	XIANSH22	51	15.2
3		WF	KUNLUN22	50	31.4
4		WF	KUNLUN22	50	31.4
5		WF	SUQIAO22	100	25.7
6	Guangxi	WF	KAIGE21	50	15.6
7		WF	DUJIG21	100	100
8		WF	CHUYUE22	25	10.0
9		WF	CHUYUE22	144	15.3
10		WF	BANGPU21	100	35.7
11		WF	DANYA22	100	10.0
12		PV	DONGTANG	245	90.0
13		PV	XIANLING	163	15.0
14	Guangdong	WF	WENTAO	198	108
15		WF	BAJITOU	297	15.0
16		WF	PINGDI	200	10.0
17		WF	PINGDI	100	10.0
18		WF	4CHEJ21	160	110
19		WF	4CHEJ21	50	10.0
20		WF	4GANT21	149	10.0
21		WF	4JIUZ21	50	10.0
22	Guizhou	WF	4ZHONH22	99	80
23		WF	4HEZ21	50	10
24		WF	4WEIN22	100	10
25		WF	4WEIN22	100	10
26		WF	4YOUY21	100	12
27		WF	WS222	50	16
28		WF	SMB22	144	70
29		WF	QSH22	49.5	20
30		WF	EY22	62.5	24
31		WF	SLB22	49.5	20
32		WF	YIC22	49.5	16
33		WF	DL22	49.5	16
34	Yunnan	WF	HPB22	99	10
35		WF	HPB22	132	40
36		WF	LUCH22 T	48	20
37		WF	HP22	92	30
38		WF	ZIX221	150	80
39		WF	ZIX221	160	110
40		WF	LF22	125	110
41		WF	LF22	98	20
42		WF	XY22X	125	32

maximum fluctuation multiples of $P_{\mathrm{RE}i0}$. Given fluctuation multiple intervals are [0.9, 1.1], [0.75, 1.25], and [0.5, 1.5], but upper bound of RES output fluctuation interval cannot exceed rated capacity.

The proposed TLOPF, MCS, LHS and AA method are used to calculate SVSM interval of the system. Since MCS and LHS methods must sample random variables of 42 RES outputs, numerous samples are needed to cover possible scenarios of random variables. There is an exponential growth relationship between number of samples and number of RESs. Even if each random variable is sampled only 10 times within its fluctuation range, 10^{42} total samples will be required. SVSM value of each sample is calculated by solving the deterministic bi-level OPF model in Section 2.1 10^{42} times, leading to great computational burden. To reduce computational burden of the two sampling methods, assume uncertain fluctuation multiples of all RES outputs in one provincial power grid of CSPG are the same. That is, all RES outputs in one provincial power grid are simplified to a random variable for sampling. MCS method samples 10 times in multiple RES output fluctuation intervals of one provincial power grid, and total sampling number is simplified to be 10^4 times. LHS method divides value space of four random variables into 2000 subspaces, and independent sampling is carried out in each subspace. Finally, 2000 RES output samples are randomly generated. Calculation results of TLOPF, MCS, LHS and AA methods are compared in Table X. Worst-case SVSM intervals obtained by the four methods are close, indicating even in an actual large AC/DC power systems, the proposed TLOPF method has high computational accuracy. Moreover, worst-case SVSM intervals calculated by the proposed TLOPF method are slightly wider than results of the two sampling methods. This is mainly because the large number of random variables means finite sampling cannot cover all possible scenarios of random variable fluctuations. It is difficult to accurately identify worst-case SVSM interval by the two sampling methods. In addition, obtained worst-case SVSM interval of AA method has obviously bigger errors than of the proposed TLOPF method with results of two sampling methods as the benchmark, especially in the case fluctuation ranges of the RES outputs are relatively big.

TABLE X
COMPARISON OF THE OBTAINED WORST-CASE SVSM INTERVALS BY FOUR METHODS

$\boldsymbol{P}_{\mathrm{RE}}$ interval	Method	SVSM interval (MW)	CPU time (s)
$[0.90, 1.10] * P_{\mathrm{RE}i0}$	TLOPF	[13553.2, 14028.1]	996.7
	MCS	[13621.2, 13948.7]	105492.3
	LHS	[13600.1, 13981.0]	21604.3
	AA	[13460.3, 14156.5]	835.4
$[0.75, 1.25] * P_{\mathrm{RE}i0}$	TLOPF	[13368.4, 14392.7]	1082.5
	MCS	[13416.1, 14274.0]	115146.1
	LHS	[13393.8, 14329.6]	22329.2
	AA	[13228.2, 14701.9]	862.0
$[0.50, 1.50] * P_{\mathrm{RE}i0}$	TLOPF	[13106.8, 14938.7]	1165.9
	MCS	[13165.1, 14847.6]	123612.2
	LHS	[13137.5, 14898.1]	23422.4
	AA	[12880.1, 15084.8]	878.6

By comparing consumed CPU time of the four methods, consumed time of the proposed TLOPF method is much less than of the other two methods at only about 1000 s. For an actual AC/DC power system with multiple RESs, MCS and LHS methods for finite sampling of multiple random variables cannot cover all possible scenarios of uncertain RES output fluctuation. These two methods not only perform poorly in obtaining accurate upper and lower boundary values of worst-case SVSM interval but also exhibit low computational efficiency. CPU time of AA method is a little faster than of the proposed TLOPF method, which is less than 900 s. However, RES outputs corresponding to the two bounds of worst-case SVSM interval cannot be directly obtained by AA method. Thus, the proposed TLOPF method performs better in worst-case SVSM interval identification of large-scale AC/DC power systems than the two sampling methods and the AA method.

Due to a large number of RESs, only fluctuation multiples of $P_{\mathrm{RE}0}$ of each bus corresponding to two bounds of the worst-case SVSM interval when given fluctuation multiple interval is [0.75, 1.25] are shown in Fig. 6. It can be observed for most RESs in the Guangdong province, decrease of RES output will increase worst-case SVSM, such as RES number 12–13 and 15–16; while for most RESs in the Guizhou province,

increase of RES output will increase worst-case SVSM, such as RES number 18–24. Thus, obtained RES output values corresponding to bounds of worst-case SVSM can supply some useful information about how to increase system SVSM by adjusting different RES outputs.

When fluctuation multiple interval of RES output is [0.75, 1.25] and worst-case SVSM reaches its lower bound, changes in inverter station variables of the LCC/VSC-HVDC transmission lines are shown in Figs. 7 and 8, respectively. Fig. 7 shows the changing process wherein LCC inverter stations of Tian-Guang HVDC and Jin-Zhong HVDC switch to constant extinction angle control when extinction angles of LCC inverter stations reach lower limit due to load growth. When ξ_i decreases to 8°, control mode is switched to constant extinction angle control, and then $U_{\mathrm{d}i}$ cannot remain constant and decreases with load growth.

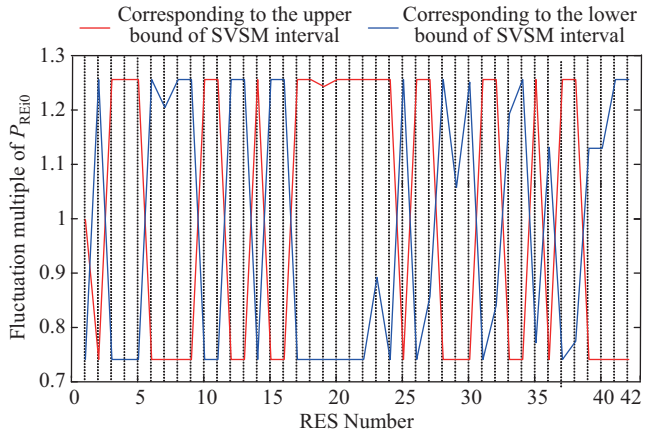


Fig. 6. Fluctuation multiple of P_{REi0} corresponding to the two bounds of the worst-case SVSM.

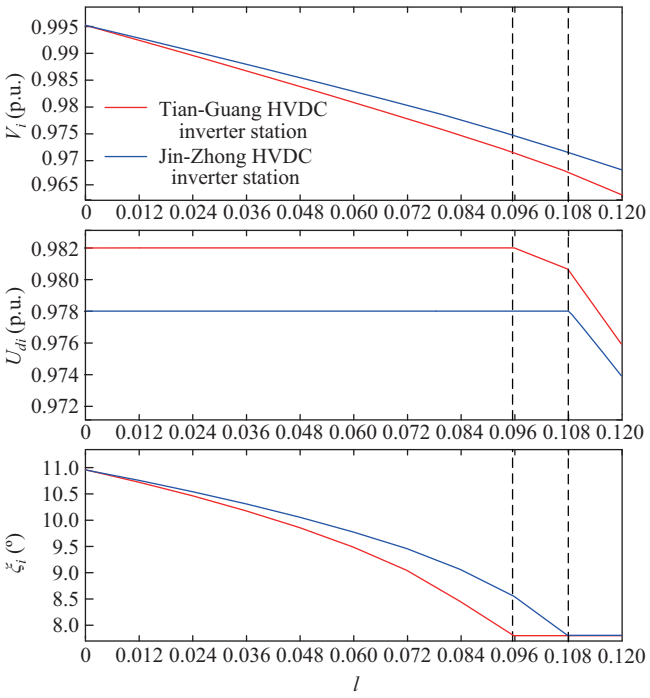


Fig. 7. Change of LCC-HVDC inverter station variables with load growth.

Figure 8 shows the changing process wherein VSC inverter station Longmen switches to constant AC voltage control when AC voltage of VSC inverter station reaches lower limit due to load growth. When V_i decreases to 0.95 p.u., control mode is switched to constant V_i control, and then $Q_{\mathrm{dc}i}$ cannot remain constant and decreases with load growth. Moreover, Figs. 7 and 8 show in a large-scale power system, since there are more HVDC transmission lines, control mode switching of the HVDC inverter station is more common than in a small-scale power system. Therefore, in the proposed the worst-case SVSM calculation model for an AC/DC power system, using (13) and (14) to describe control mode switching of DC converter station with load growth is necessary and effective.

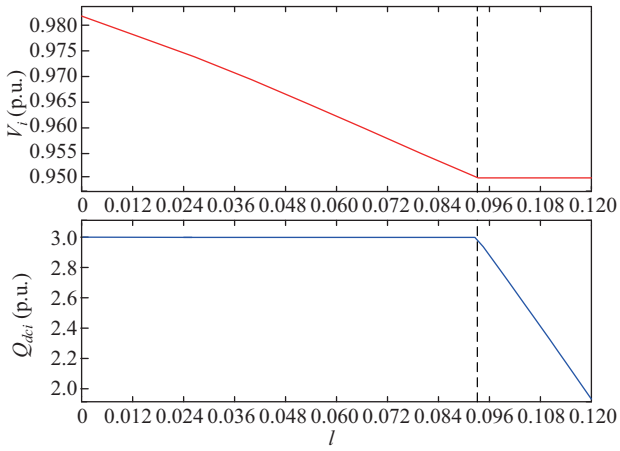


Fig. 8. Change of VSC-HVDC inverter station variables with load growth.

V. Conclusion

In this study, an identification method for worst-case SVSM interval of AC/DC power systems considering uncertain RES output is proposed. The proposed method uses a hypercone model to describe LGD uncertainty. A bi-level identification model is established to solve worst-case SVSM. Two TLOPF models are proposed to identify worst-case SVSM interval with uncertain RES output. Control mode switching of DC converter station along with load growth is considered in calculation models. The two TLOPF models can both be transformed into max–min bi-level OPF models at first. By transforming inner level model into its dual form, bi-level max–min OPF models can be transformed into two single-level optimization models for solution. Case studies on modified IEEE-39 bus AC/DC system and an actual AC/DC power grid in China indicate correctness and efficiency of the proposed method. Moreover, obtained worst-case SVSM intervals are more conservative than traditional SVSM intervals under the initial LGD.

This study only calculates worst-case SVSM intervals of AC/DC power systems considering uncertainties of LGD and renewables. In actual operation of an AC/DC power grid, worst-case SVSM value must be bigger than allowable secure level. Thus, our future approach will be to increase worst-case SVSM value by adjusting system operation parameters

and implementing some control measures, especially by establishing a new control mode of HVDC transmission system with some methods.

REFERENCES

[1] J. Lu, X. M. Yuan, J. B. Hu, M. Q. Zhang, and H. Yuan, "Motion equation modeling of LCC-HVDC stations for analyzing DC and AC network interactions," *IEEE Transactions on Power Delivery*, vol. 35, no. 3, pp. 1563–1574, Jun. 2020.

[2] F. Zhang, H. H. Xin, D. Wu, Z. Wang, and D. Q. Gan, "Assessing strength of multi-infeed LCC-HVDC systems using generalized short-circuit ratio," *IEEE Transactions on Power Systems*, vol. 34, no. 1, pp. 467–480, Jan. 2019.

[3] L. M. Castro and E. Acha, "A unified modeling approach of multi-terminal VSC-HVDC links for dynamic simulations of large-scale power systems," *IEEE Transactions on Power Systems*, vol. 31, no. 6, pp. 5051–5060, Nov. 2016.

[4] B. R. Zhou, H. Rao, W. Wu, T. Wang, C. Hong, D. Q. Huang, W. F. Yao, X. R. Su, and T. Mao, "Principle and application of asynchronous operation of China southern power grid," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 6, no. 3, pp. 1032–1040, Sep. 2018.

[5] Y. R. Yang, S. J. Lin, Q. Wang, Y. Q. Xie, M. B. Liu, and Q. F. Li, "Optimization of static voltage stability margin considering uncertainties of wind power generation," *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4525–4540, Nov. 2022.

[6] Q. Wang, S. J. Lin, Y. R. Yang, and M. B. Liu, "A decomposition and coordination algorithm for SVSM interval of integrated transmission and distribution networks considering the uncertainty of renewable energy," *International Journal of Electrical Power & Energy Systems*, vol. 136, pp. 107761, Mar. 2022.

[7] J. Zhao, Y. Wang, and P. Xu, "A comprehensive on-line voltage stability assessment method based on continuation power flow," in *2009 International Conference on Sustainable Power Generation and Supply*, 2009, pp. 1–5.

[8] C. Y. Guo, Y. Zhang, A. M. Gole, and C. Y. Zhao, "Analysis of dual-Infeed HVDC with LCC–HVDC and VSC–HVDC," *IEEE Transactions on Power Delivery*, vol. 27, no. 3, pp. 1529–1537, Jul. 2012.

[9] D. H. A. Lee and G. Andersson, "An equivalent single-infeed model of multi-infeed HVDC systems for voltage and power stability analysis," *IEEE Transactions on Power Delivery*, vol. 31, no. 1, pp. 303–312, Feb. 2016.

[10] D. D. Li, M. X. Sun, and Y. Fu, "A general steady-state voltage stability analysis for hybrid multi-infeed HVDC systems," *IEEE Transactions on Power Delivery*, vol. 36, no. 3, pp. 1302–1312, Jun. 2021.

[11] O. A. Urquidez and L. Xie, "Singular value sensitivity based optimal control of embedded VSC-HVDC for steady-state voltage stability enhancement," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 216–225, Jan. 2016.

[12] S. Mirsaeidi, X. Z. Dong, D. Tzelepis, D. M. Said, A. Dyśko, and C. Booth, "A predictive control strategy for mitigation of commutation failure in LCC-Based HVDC systems," *IEEE Transactions on Power Electronics*, vol. 34, no. 1, pp. 160–172, Jan. 2019.

[13] W. J. Du, Q. Fu, and H. F. Wang, "Comparing AC dynamic transients propagated through VSC HVDC connection with master–slave control versus DC voltage droop control," *IEEE Transactions on Sustainable Energy*, vol. 9, no. 3, pp. 1285–1297, Jul. 2018.

[14] S. J. Lin, Y. Lu, M. B. Liu, Y. R. Yang, S. He, and H. Jiang, "SVSM calculation of power system with high wind-power penetration," *IET Renewable Power Generation*, vol. 13, no. 8, pp. 1391–1401, Jun. 2019.

[15] A. Vaccaro, C. A. Canizares, and D. Villacci, "An affine arithmetic-based methodology for reliable power flow analysis in the presence of data uncertainty," *IEEE Transactions on Power Systems*, vol. 25, no. 2, pp. 624–632, May 2010.

[16] J. Muñoz, C. Cañizares, K. Bhattacharya, and A. Vaccaro, "An affine arithmetic-based method for voltage stability assessment of power systems with intermittent generation sources," *IEEE Transactions on Power Systems*, vol. 28, no. 4, pp. 4475–4487, Nov. 2013.

[17] S. J. Lin, Y. R. Yang, M. B. Liu, Y. Q. Xie, and Y. Lu, "Static voltage stability margin calculation of power systems with high wind power penetration based on the interval optimisation method," *IET Renewable Power Generation*, vol. 14, no. 10, pp. 1728–1737, Jul. 2020.

[18] X. Li, L. W. Zhang, T. Jiang, F. X. Li, H. H. Chen, and H. J. Jia, Relaxed decoupled direct calculation of voltage collapse points and its application in static voltage stability region boundary formation," *International Journal of Electrical Power & Energy Systems*, vol. 125, pp. 106452, Feb. 2021.

[19] I. Dobson and L. Lu, "New methods for computing a closest saddle node bifurcation and worst case load power margin for voltage collapse," *IEEE Transactions on Power Systems*, vol. 8, no. 3, pp. 905–913, Aug. 1993.

[20] F. Alvarado, I. Dobson, and Y. Hu, "Computation of closest bifurcations in power systems," *IEEE Transactions on Power Systems*, vol. 9, no. 2, pp. 918–928, May 1994.

[21] Y. Kataoka, "A probabilistic nodal loading model and worst case solutions for electric power system voltage stability assessment," *IEEE Transactions on Power Systems*, vol. 18, no. 4, pp. 1507–1514, Nov. 2003.

[22] Y. Z. Sun, L. Peng, F. Ma, G. J. Li, and P. F. Lv, "Design a fuzzy controller to minimize the effect of HVDC commutation failure on power system," *IEEE Transactions on Power Systems*, vol. 23, no. 1, pp. 100–107, Feb. 2008.

[23] W. Y. Wang, A. Beddard, M. Barnes, and O. Marjanovic, "Analysis of active power control for VSC–HVDC," *IEEE Transactions on Power Delivery*, vol. 29, no. 4, pp. 1978–1988, Aug. 2014.

[24] W. N. Li and Y. H. Chen, "MISO AGC enhancement proposal to better utilize fast ramping resources," in *2015 IEEE Power & Energy Society General Meeting*, 2015, pp. 1–5.

[25] C. H. Peng, P. Xie, L. Pan, and R. Yu, "Flexible robust optimization dispatch for hybrid wind/photovoltaic/hydro/thermal power system," *IEEE Transactions on Smart Grid*, vol. 7, no. 2, pp. 751–762, Mar. 2016.

[26] C. Liu, C. Lee, H. Y. Chen, and S. Mehrotra, "Stochastic robust mathematical programming model for power system optimization," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 821–822, Jan. 2016.

[27] J. Y. Zhai, M. Zhou, J. N. Li, X. P. Zhang, G. Y. Li, C. Y. X. Ni, and W. Y. Zhang, "Hierarchical and robust scheduling approach for VSC-MTDC meshed AC/DC grid with high share of wind power," *IEEE Transactions on Power Systems*, vol. 36, no. 1, pp. 793–805, Jan. 2021.

[28] P. Huard, "Dual programs," *IBM Journal of Research and Development*, vol. 6, no. 1, pp. 137–139, Jan. 1962.

[29] Q. F. Li and V. Vittal, "Convex hull of the quadratic branch AC power flow equations and its application in radial distribution networks," *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 839–850, Jan. 2018.

[30] G. P. McCormick, "Computability of global solutions to factorable nonconvex programs: part I—Convex underestimating problems," *Mathematical Programming*, vol. 10, no. 1, pp. 147–175, Dec. 1976.

[31] A. Drud. (2022, Feb.). CONOPT. [Online]. Available: https://www.gams.com/latest/docs/S_CONOPT.html.

[32] A. A. Yusuff, "Voltage stability index based on standard deviation-mean ratio for identification of weak nodes," in *2017 IEEE AFRICON*, 2017, pp. 1255–1259.

[33] W. K. Liang, S. J. Lin, J. Liu, P. Dong, and M. B. Liu, "Multi-objective mixed-integer convex programming method for the siting and sizing of UPFCs in a large-scale AC/DC transmission system," *IEEE Transactions on Power Systems*, Nov. 2022, doi: 10.1109/TPWRS.2022.3220566.

[34] L. Xiong, X. Liu, Y. Liu and F. Zhuo, "Modeling and Stability Issues of Voltage-source Converter-dominated Power Systems: A Review," *CSEE Journal of Power and Energy Systems*, vol. 8, no. 6, pp. 1530–1549, Nov. 2022.

[35] P. Sun, Y. Teng, O. Leng and Z. Chen, "Stability control method for hybrid AC-DC transmission systems considering cross-region multi-energy coordination," *CSEE Journal of Power and Energy Systems*, vol. 7, no. 4, pp. 753–760, Jul. 2021.



Wanbin Liu received a B.S. degree in Electrical Engineering from the South China University of Technology in 2021, where he is currently pursuing his M.S. degree in Electrical Engineering. His research interests include power system optimization, operation and control.



Shunjiang Lin received a B.S. degree in Electrical Engineering from South China University of Technology, in 2003, and a Ph.D. degree in Electrical Engineering from Hunan University, in 2008. He is currently an Associate Professor with the School of Electric Power Engineering, South China University of Technology. His research interests include power system optimization, operation and control.



Yuerong Yang received a B.S. degree in Electrical Engineering from the South China University of Technology, in 2019. He is currently pursuing an M.S. degree with the School of Electric Power Engineering, South China University of Technology. His research interests include power system optimization, operation, and control.



Mingbo Liu received a B.S. degree from the Huazhong University of Science and Technology, in 1985, an M.S. degree from the Harbin Institute of Technology, in 1988, and a Ph.D. degree from Tsinghua University, in 1992. He is currently a Professor with the South China University of Technology. He has authored and coauthored four monographs, two standards, and more than 280 articles. His research interests include energy management and operation control of power systems.



Qifeng Li received a Ph.D. degree in Electrical Engineering from Arizona State University, Tempe, AZ, USA, in 2016. He is currently an Assistant Professor with the Department of Electrical and Computer Engineering, University of Central Florida (UCF), Orlando, FL, USA. Before joining UCF Faculty, he held a position of a Postdoctoral Associate with the Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA, USA, from 2016 to 2018. His research interests include convex optimization, uncertainty-aware optimization, and nonlinear systems with applications in power and energy systems.